Ans to Ex.1

1. .(a) Table of expost forecasts:

Data	Naïve	3-point MA	Trend	Causal
1435.9	1365.8	1271.8	1366.7	1444.15
1521.4	1435.9	1361.1	1433	1524.33
1629.2	1521.4	1441	1499.29	1653.13
1733.7	1629.2	1528.8	1565.59	1763.04
1807.2	1733.7	1628.1	1631.89	1871.59

Note: Trend model and Causal model are based on observations between 1990 to 2005.

Trend equation : $y_t = 239.63+66.3*t$
Regression eqn.: $y_t=78.7+0.454x_t$

(b) & (c)

Obviously the casual model is the best from the expost forecasts.

2. (a), (b), & (c)

Year	Data	Naïve	MA	Trend	Causal
98	0.63	0.62	0.59	1.023	0.931
99	0.86	0.63	0.61	1.024	0.887
00	1.19	0.86	0.70	1.025	1.146
01	1.31	1.19	0.89	1.026	1.325
02	1.22	1.31	1.12	1.027	1.241
03	1.16	1.22	1.24	1.028	1.169
04	1.13	1.16	1.23	1.029	1.16
05	1.12	1.13	1.17	1.030	1.113
06	0.86	1.12	1.14	1.031	0.808
07	0.90	0.86	1.04	1.032	0.892
08	0.90	0.90	0.96	1.033	0.812

Note: Time has been recoded, namely, 1998 to 2008 has been recoded as 1 to 11. Also the slope of the trend equation given is not accurate, but in the answer above I still use the given equation.

(d)

	Naïve	MA	Trend	causal
MSE	0.023	0.055	0.039	0.0096
RMSE	0.153	0.234	0.197	0.098
MAE	0.107	0.181	0.178	0.055
MAPE	10.55%	17.24%	19.14%	7.11%

Causal model is best from all criteria.

(e)

Year	Naïve	MA	Trend
09	0.90	0.89	1.034
10	0.90	0.9	1.035

3. (a), (b) & (c).

Year	Data	Naïve	MA	Trend	Causal
98	.57	.53	.53	.718	.768
99	.72	.57	.54	.742	.717
00	.87	.72	.61	.766	.811
01	.87	.87	.72	.790	.860
02	.87	.87	.82	.814	.906
03	.87	.87	.87	.838	.878
04	.87	.87	.87	.862	.848
05	.87	.87	.87	.886	.881
06	.87	.87	.87	.910	.845
07	.90	.87	.87	.934	.768

(d) Naïve is best by all measures.

(e) Expost forecasts

Year	Naïve	MA	Trend	causal
08	0.9	0.88	0.958	0.717
09	1.15	0.97	0.982	0.703
10	1.2	1.08	1.006	0.791

(f) & (g)

Naïve forecast is the 'best' by all criteria. The reason is due to the constancy of the dividend payment over the period.

4. (a) No, as mean depends on t.
(b) Yes, as $E(Z_t)=7-3t+3t=7$, a constant.

5. $E(Y_t) = E(A) + E(X_t) = \mu_A + \mu_X$, free of t.

$Cov(Y_t, Y_{t-k}) = cov(A + X_t, A + X_{t-k})$
$= cov(A, A+X_{t-k},) + cov (X_t, A+ X_{t-k})$
$= var(A) + cov(X_t, X_{t-k})$, free of t as X_t is stationary.

6. $Y_t = \theta_0 + Y_{t-1} + e_t = \theta_0 + \theta_0 + Y_{t-2} + e_{t-1} + e_t$
$= 2\theta_0 + Y_{t-2} + e_{t-1} + e_t$
$= \ldots$
$= (t-1)\theta_0 + Y_1 + e_t + e_{t-1} + \ldots + e_2$
$= t\theta_0 + e_t + \ldots + e_1$

$E(Y_t) = t\theta_0$, dependent on t

$$\begin{aligned}\operatorname{cov}(Y_t, Y_{t-k}) &= \operatorname{cov}(t\theta_0 + \sum_{i=1}^{t} e_i, \quad (t-k)\theta_0 + \sum_{i=1}^{t-k} e_i) \\ &= (t-k)\sigma_e^{\,2} \text{ , dependent on t}\end{aligned}$$